

matreex: Simulating European forest dynamics with IPM.

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Summary

Integral projection models (IPMs) are powerful tools for studying the temporal dynamics of populations structured by continuous traits, allowing for predictions of changes in trait distributions over time ([Ellner et al., 2016](#)). Unlike individual-based or cohort-based models, which represent populations as discrete populations, IPMs describe populations as continuous populations, integrating over the uncertainty of demographic processes. This removes demographic stochasticity and results in fully deterministic simulations, which are complementary to individual-based models (IBMs).

Here, we introduce matreex, an R package specifically designed to build IPMs for European forest tree species. Our package includes pre-fitted species-specific growth, survival and recruitment functions that account for the effect of climate and competition, and functions to efficiently integrate IPMs and run temporal simulations of single-species or multispecies forest communities until equilibrium. This package complements existing R packages for IPMs, such as *ipmr* ([Metcalf et al., 2013](#)) and *IPMpack* ([Levin et al., 2021](#)), which are not specifically designed for forest ecosystems.

Statement of need

IPMs are rarely applied to forest ecosystems due to the complexity of tree growth kernels, which are challenging to integrate, making the construction of forest IPMs particularly difficult. matreex is an R package specifically designed to build IPMs and run simulations for European forests. These IPMs can be used by the forest ecology research community, to study population dynamic and forest communities trajectories. In matreex IPM simulations, it is also possible to include temporally variable climatic conditions, natural disturbances, harvesting scenarios and regional dispersal affecting population dynaminc depending on tree species sensitivity and stand structure.

State of the field

Other, more generalist, packages exist to build IPMs for various types of organisms (*IPMpack* ([Metcalf et al., 2013](#)), *ipmr* ([Levin et al., 2021](#))). However they focus on analysis on the matrix itself whereas matreex use the integrated matrices to simulate population dynamics. In addition matreex integrates numerous functions to run simulations for multispecies communities under various harvesting and disturbance scenarios.

39 Software design

40 A key feature of the `matreex` package is the development of IPM integration functions designed
41 for complex forest tree growth kernels. As trees have a small annual growth rate, most of
42 the integration effort is concentrated near the diagonal of the matrix. We achieve this by
43 combining different integration methods (Gauss-Legendre and Mid-bin) at different distances
44 of the diagonal, which helps speed up the integration. [Figure 1](#) illustrates how these different
45 methods of integrating the matrix are used. This speed is crucial as we integrate a matrix
46 per competition level (based on species basal area) to account for density dependence. More
47 details about the integration are provided in the [Building IPM vignette](#).

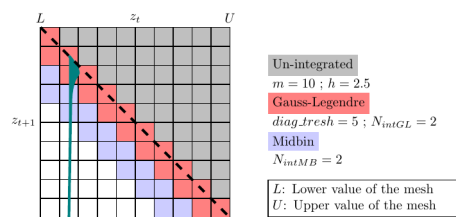


Figure 1: Combination of different integration methods. Dashed line is the identity where $z_t = z_t + 1$ and dark blue distribution is an expected distribution of the growth kernel.

48 A crucial development objective was to simplify the workflow for ecological researchers, who
49 may work on multispecies models with climate variation, disturbances, and harvesting. To
50 facilitate this, `matreex` provides fitted vital models for European tree species directly in the
51 package ([Barrere et al., 2024](#); [Guyennon et al., 2023](#); [Kunstler et al., 2021](#)), although other
52 new vital models can be used. The object-oriented architecture limits code complexity, allowing
53 users to focus on designing large simulation experiments to tackle their ecological questions.
54 The code below demonstrates a quick way to use the package, resulting in [Figure 2](#), which
55 shows the dynamics of four species in the same environment.

```
56 library(matreex)
57 library(dplyr)
58 library(ggplot2)
59
60 # select a climate to run in
61 data("climate_species")
62 climate <- subset(
63   climate_species,
64   climate_label == "optimum" & sp == "Fagus_sylvatica",
65   select = -c(sp, climate_label)
66 )
67
68 # integrate ipms and store them in species object
69 Picea <- species(IPM = make_IPM(
70   species = "Picea_abies", fit = fit_Picea_abies,
71   climate = climate, clim_lab = "optimum clim",
72   mesh = c(m = 700, L = 90, U = get_maxdbh(fit_Picea_abies) * 1.1),
73   BA = 0:60
74 ), init_pop = def_initBA(1))
75 Betula <- species(IPM = make_IPM(
76   species = "Betula", fit = fit_Betula,
77   climate = climate, clim_lab = "optimum clim",
78   mesh = c(m = 700, L = 90, U = get_maxdbh(fit_Betula) * 1.1),
79   BA = 0:60
```

```

80 ), init_pop = def_initBA(1))
81 Fagus <- species(IPM = make_IPM(
82   species = "Fagus_sylvatica", fit = fit_Fagus_sylvatica,
83   climate = climate, clim_lab = "optimum clim",
84   mesh = c(m = 700, L = 90, U = get_maxdbh(fit_Fagus_sylvatica) * 1.1),
85   BA = 0:60
86 ), init_pop = def_initBA(1))
87 Abies <- species(IPM = make_IPM(
88   species = "Abies_alba", fit = fit_Abies_alba,
89   climate = climate, clim_lab = "optimum clim",
90   mesh = c(m = 700, L = 90, U = get_maxdbh(fit_Abies_alba) * 1.1),
91   BA = 0:60
92 ), init_pop = def_initBA(1))
93
94 # assemble species in a forest object
95 Forest <- forest(species = list(Picea = Picea, Abies = Abies,
96                               Fagus = Fagus, Betula = Betula))
97 set.seed(20260825) # The seed is here for initial population random functions.
98
99 # Run simulation and plot it
100 Sim <- sim_deter_forest(
101   Forest,
102   tlim = 1000,
103   equil_time = 1000, equil_dist = 50, equil_diff = 1,
104   SurfEch = 0.03,
105   verbose = TRUE
106 )
107 Sim %>%
108   dplyr::filter(var == "BAsp", ! equil) |>
109   ggplot(aes(x = time, y = value, color = species)) +
110   geom_line(linewidth = .4) +
111   ylab("Basal Area (m2)") + xlab("Simulation time (years)")

```

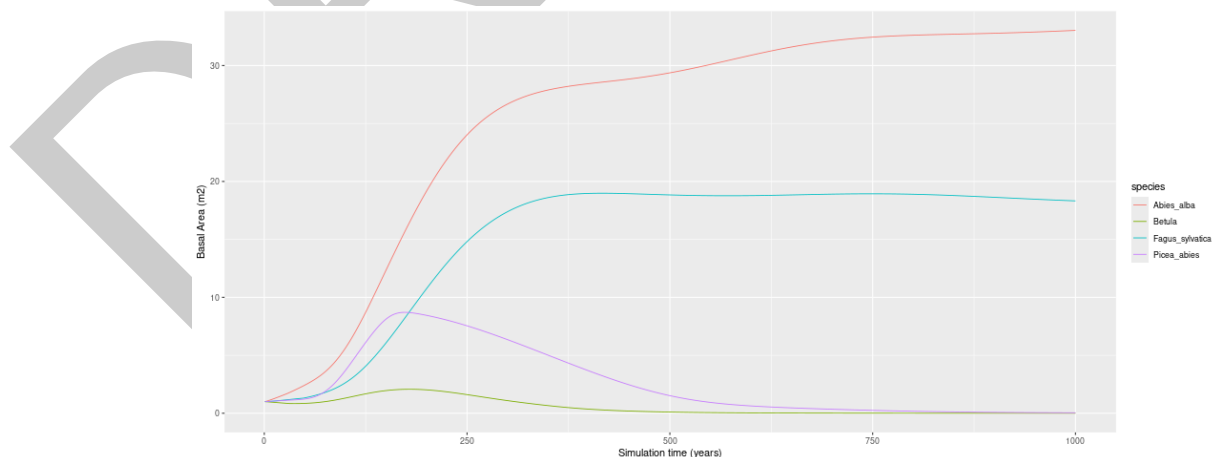


Figure 2: Simulation output for 4 species.

112 Climatic temporal variability

113 Modelling forest dynamics under fluctuating climatic conditions can be computationally
 114 expensive because the IPMs growth kernel must be integrated for every climatic condition. To

115 avoid this high computation cost, we implement a new method involving the pre-integration
116 of IPM growth matrix blocks for different mean growth rates. The IPM for each climatic
117 condition is then reassembled from these mean growth rate IPM matrix blocks (mu matrix).
118 This allows to speed-up the simulations. This is described in the [matreex climate variation](#)
119 [vignette](#).

120 Disturbance

121 One key originality of matreex is the possibility to apply storm, fire, biotic or snow disturbances
122 of varying intensity (ranging from 0 to 1) at any time of the IPM simulations. When a
123 disturbance strikes the stand a given year of the simulation, the survival function is replaced
124 by the species-specific equations from Barrere et al. (2024). These equations quantify the
125 annual mortality probability of a tree in a disturbed stand as a function of its species, diameter
126 at breast height, stand structure, nature and intensity of the disturbance. A disturbance
127 striking two different stands with the same intensity will thus result in different mortality rates,
128 depending notably on the sensitivity of the tree species present in the stand. Disturbances in
129 matreex are described in details in Barrere et al. (2024) and in Barrere et al. (in prep). Figure
130 3 shows an example of multispecies simulations with storm disturbance.

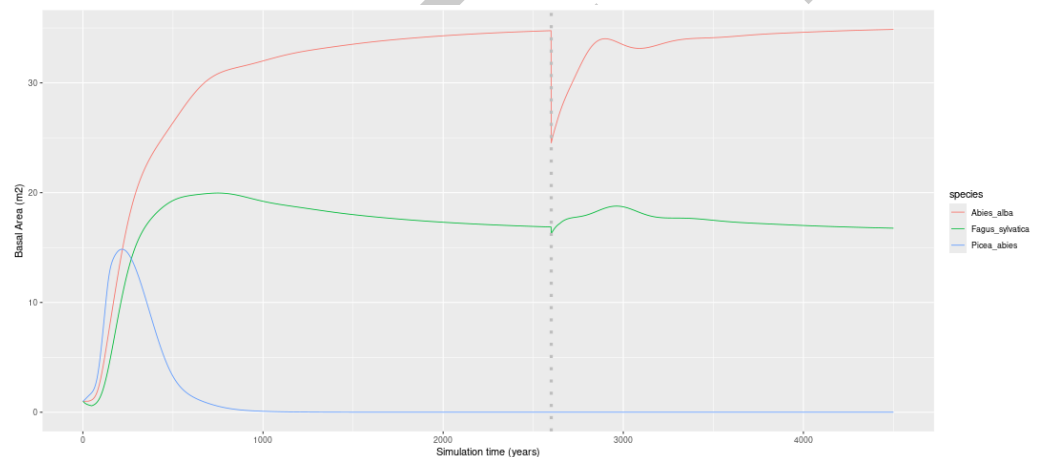


Figure 3: Simulation output for 3 species with a disturbance at *time* = 2600.

131 Regional dispersal

132 Most stand-scale forest dynamics models simulate closed systems, where only the tree species
133 already present in the stand contribute to the recruitment of new trees. This limitation preclude
134 the simulation of immigration from external species, which is a key process of forest dynamics,
135 particularly under climate change. To overcome this limitation, we included the possibility
136 to split the recruitment function in two component : (i) within-plot dispersal that depends
137 on the sum of basal area of fecund tree species in the plot, and (ii) external dispersal that
138 depends on the regional pool. This regional pool approach is extensively presented in Barrere
139 et al. (in prep).

140 Harvesting

141 Since most temperate forests are managed, it is crucial to incorporate silvicultural interventions
142 into the simulations. We implemented three management strategies:

- 143 ■ First, we implemented a simple constant annual harvesting rates accounting for the effect
144 of the harvesting rates observed in NFI data used for the model calibration ([Kunstler et](#)
145 [al., 2021](#)).

- 146 ▪ Second, we implement an even-aged management. The objective is to apply typical
147 even-aged harvesting, based on a single cohort. Trees are harvested with successive
148 thinning during stand development until the final harvest. Thinning harvest are based
149 on the distance to a self-thinning boundary, based on Aussenac et al. (2021). This is
150 easily connected with management guidelines.
- 151 ▪ Third, we implement an unven-aged harvesting. The uneven-aged harvest scenario
152 consists in selective harvesting across all size classes with the objective to reach a stable
153 size structure with continuous replacement of large mature trees. This scenario depends
154 on the basal area of the stand and the size distribution of the trees (building on Lafond et
155 al. (2014)). These three managements are described in the [matreex harvesting vignette](#).

156 Research impact statement

157 matreex was designed to be easily adapted to various research ideas and is in continuous
158 development. It has already been used in several scientific publications tackling diverse
159 scientific questions (Baranger et al., in prep; Barrere et al., 2024, in prep; Guyennon et al.,
160 2023; Kunstler et al., 2021). The ability to simulate forests easily with a dedicated R package
161 will help ecologists to analyse the effect of climate change, shifting disturbance regimes, and
162 their interplay with forest management across European forests.

163 matreex is an open-source package made available under the MIT license. Installation and
164 usage instructions can be found at the [website](#)

165 AI usage disclosure

166 No generative AI tools were used in the development of this software, the writing of this
167 manuscript, or the preparation of supporting materials.

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